Forensic red-team F1: $h^{\text{readout}}(z_{\text{probe}})$ under the v4.5 two-tier coherence criterion

(retraction dissection; new-claim hunt; Candidate 3 derivation attempt)

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Abstract

V2.2 of the Physics Ledger retracted the v1 "orbit-depth-dependent readout" narrative that had motivated τ's $h = 0.6735$ surviving alongside local-ladder $h \approx 0.73$. Three derivation attempts (Approaches A, B, C in the v3 scoping note) failed. V4.5 Ramanathan subsequently identified a two-tier coherence criterion on $H_\partial[\omega]$: global cosmic-mean densities sit at *tier 1* (D-sector holonomy ratio $(1-\iota_\tau)/\iota_\tau^2 \approx 6.65\times$), while local matter-phase couplings sit at *tier 2* (D/ω/B trinary with weights $\{1, \kappa_\omega, \approx 0\}$). Post-v4.5, this note dissects each of the three failed derivations in light of the two-tier structure, hunts for a valid new h-claim, and actually attempts the most ambitious ("Candidate 3: tier-2 probe-sector-weighting formula"). The result is a clean verdict (b): the two-tier structure *sharpens* the tension but does *not* produce a falsifiable h^{readout} prediction. Forensic reason: H_0 is a tier-1 observable (an expansion rate of the cosmic-mean background), and the probe-to-probe disagreement cannot be absorbed into tier-2 coupling weights — both *stellar-sector* probes (Cepheid: $h = 0.730$; TRGB: $h = 0.698$) live in the *same* tier-2 D-sector yet disagree at 3σ, killing any sector-weighting explanation at the kinematical level. The retraction stands; the new insight is narrative refinement for v2.5, not new physics. We propose a revised v2.5 framing that incorporates the two-tier insight without overreach.

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1 Scope and charge

The v2.2 Ledger retracted an "orbit-depth-dependent readout" claim (`rem:ch63-h0-depth-flag`) after three derivation approaches failed in the v3 scoping note (`research/v3-scoping-2026-04-20/h-readout-derivation-note.tex`). R4 (McGaugh)'s "semantic maneuver" charge was upheld; N21 was relabeled to "τ predicts $h = 0.6735$; 3–6σ tension with TRGB/H0LiCOW/SH0ES."

V4.5 Ramanathan (`research/v4-deep-research-wave-2026-04-20/v4_5_cosmology_consistency/v4-5-cosmology-consistency-result.tex`) subsequently established a *two-tier* coherence criterion on the boundary character algebra $H_\partial[\omega]$:

- **Tier 1 (global background).** Cosmic-mean densities ω_m, ω_b, ω_r, Ω_Λ are properties of the D-sector of $H_\partial[\omega]$ as a whole. The $(1-\iota_\tau)/\iota_\tau^2 \approx 6.65\times$ holonomy ratio dresses ω_m/ω_b at this level.
- **Tier 2 (local matter-phase coupling).** The trinary weights $\{1, \kappa_\omega, \approx 0\}$ for stars / cold HI / hot ICM are a *local* coupling of each matter phase into the capacity-gradient source term in the rotation-curve / BTFR / cluster-lensing equation.

The two tiers are structurally decoupled: Tier 1 is set upstream of Tier 2 by the M3h holonomy argument (V.T198) and does not propagate through the tier-2 local weights.

Forensic charge. Given the two-tier refinement, was the v1/v2.1 "depth-dependent readout" the inchoate form of a claim that the two-tier criterion now legitimises? If so, what is the valid claim, and how does it test? If not, how does the two-tier insight reshape the v2.5 narrative without overclaiming?

2 Forensic of Approaches A, B, C under the two-tier lens

We walk through each failed derivation and ask the new question: *did the two-tier criterion exist implicitly in the framework, and would the derivation have closed if it had been made explicit?*

2.1 Approach A — depth as $f(z_{\text{probe}})$

v2.2 obstruction (recap). Four probes at $z \sim 0.01$ report four different h values: $h_{\text{SH0ES}} = 0.730$, $h_{\text{sirens}} = 0.700$, $h_{\text{TRGB}} = 0.698$, $h_{\text{JAGB}} = 0.695$. A scalar function $f(z)$ is single-valued and cannot return four outputs at one input. Hence A is geometrically ruled out.

Forensic under two-tier. The two-tier structure does not rescue Approach A, because the tiers do not *encode* z_{probe}. Tier 1 is a cosmic-mean statement (no probe); Tier 2 is a galaxy/cluster-scale statement (no redshift argument). The tier assignment of a probe is a categorical attribute ("which sector does the measured mass phase live in"), not a function of probe redshift. Promoting z_{probe} to $(z_{\text{probe}}, \text{tier-2 sector})$ reduces the problem to Candidate 3 below, which we attempt as a separate derivation. The scalar-z form of A remains dead.

Verdict on A. *Retraction stands. The two-tier criterion does not refine the depth-scalar parametrisation, because the tiers are orthogonal to z.* What the two-tier insight does provide is a *reinterpretation* of the A-ansatz: the original "$f(z_{\text{probe}})$" was reaching for a sector-weighting that the framework simply does not express as a scalar function of redshift. Approach A was posing the wrong question.

2.2 Approach B — readout-functor projection

v2.2 obstruction (recap). Ch. 66's readout map (`eq:ch66-readout-map`)

$$\text{Read}\colon H_\partial[\omega] \longrightarrow \mathscr{H}_{\text{VM}}, \qquad \chi \mapsto |\psi_\chi\rangle$$

has one argument (a boundary character) and one output (a VM state); it admits no probe-depth argument and produces a probe-depth-isotropic observable.

Forensic under two-tier. Can the readout functor be factored through the two tiers, with the tier-1/tier-2 distinction supplying the missing argument?

Consider the factorisation

$$\text{Read} \;=\; \text{Read}^{(2)} \circ \pi_T \;\circ\; \iota^{(1)}_{H_\partial[\omega]} \tag{1}$$

where $\iota^{(1)}_{H_\partial[\omega]}$ embeds a background tier-1 character into $H_\partial[\omega]$ and π_T projects onto a tier-2 sector component of $T \in \{D, \omega, B\}$. This factorisation is consistent with V4.5's derivation: the D-sector background content lives at tier 1, and the trinary decomposition is a tier-2 projection on top of that.

Does π_T give the readout a probe argument? *No.* π_T indexes matter phases (stars, cold HI, hot ICM), not probes. The readout is now tier-dependent in the sense that a cosmological observable (tier 1) and a galaxy-rotation observable (tier 2) use different factorisations, but the H_0 expansion-rate observable is still a tier-1 observable (it is the expansion rate of the cosmic-mean background). Under (1), h is computed via

$$h \;=\; \langle \psi_{\chi_{\text{bg}}} \mid \hat{O}_H \mid \psi_{\chi_{\text{bg}}} \rangle, \qquad \chi_{\text{bg}} \in \iota^{(1)}_{H_\partial[\omega]}, \tag{2}$$

with no tier-2 projection entering. The tier-1 character of the cosmological background is single, hence h is single.

Proposition 1 (Two-tier isotropy)**.** *Under the factorisation* (1) *and the tier-1 assignment of ω_m, ω_b, h in V4.5, the Ch. 66 readout of $\hat{O}_H$ is tier-2-independent: $h^{(T)} = h^{(T')}\ \forall\, T, T' \in \{D, \omega, B\}$.*

Proof sketch. $\hat{O}_H$ is a functional of the background Friedmann cosmology, which is set by tier-1 densities $(\omega_m, \omega_b, \Omega_\Lambda, h)$. These are upstream of π_T (V4.5, §3.2). The readout of $\hat{O}_H$ evaluated on any tier-2 projection of the background character factors through the same tier-1 input. □

Verdict on B. *The two-tier structure does refine the readout functor, but the refinement makes the h-tension* worse *for a depth-dependent rescue, not better.* The factorisation (1) makes explicit what was implicit: h is a tier-1 observable, and tier-2 sector weights cannot reach it. The v2.2 "isotropic readout" conclusion is reaffirmed with a cleaner structural reason. Approach B is still dead, but for a *better* reason now: not "the functor's signature forbids depth" (which could have been read as incompleteness), but "the observable lives at the tier where depth cannot enter" (which is a framework-level feature).

2.3 Approach C — line-of-sight functional on $H_\partial[\omega]$

v2.2 obstruction (recap). No line-of-sight functional exists on $H_\partial[\omega]$; the boundary character algebra is defined globally, not along geodesics (Lemma on no-path-functional in v3 note).

Forensic under two-tier. A line-of-sight functional would, in the two-tier picture, accumulate contributions from the tier-2 local sector of the matter intersected along the geodesic. Example: a strong-lens time delay (H0LiCOW, $z \sim 0.5$) is a line-of-sight integral through a cluster's mass distribution, which contains a mix of stars (D-sector), gas (ω-sector), and the lensing deflection itself probes the D-sector background (V4.5 §5 cluster argument).

Does this give a valid functional? *Partially.* A consistent *line-of-sight accumulation* of tier-2 weights along a geodesic would be a well-defined object, *if* the object being accumulated were a local tier-2 source contribution to the H_0 inference. But H_0 is *not* inferred from local tier-2 sources in the standard distance-redshift relation. Cepheid-SN ladders infer H_0 via

$$h \;=\; \frac{v_{\text{recession}}}{c \cdot d_L(z)},$$

where d_L is the luminosity distance. The ladder calibrates d_L via stellar pulsation physics (Cepheid P-L relation), not via a tier-2 coupling of the Cepheid into the host galaxy's capacity-gradient source. The pulsation physics is *orthodox stellar physics*, not a tier-2 coupling weight.

Lemma 1 (H_0 ladders are tier-1 probes of Friedmann $H(z)$, not tier-2 probes of local sector weights)**.** *The standard distance-ladder inference of H_0 from a probe yielding (z, d_L) tests the Friedmann expansion-rate profile $H(z)$, which is a tier-1 observable. Tier-2 sector weights enter neither the luminosity-distance integrand $\int dz/H(z)$ nor the calibration of d_L via stellar physics.*

Sketch. (i) Luminosity distance $d_L(z) = (1+z)\int_0^z c\,dz'/H(z')$ depends on $H(z)$ which is set by tier-1 densities; (ii) the calibration of d_L at anchor distances (LMC for Cepheid, NGC 4258 for megamasers) uses geometric parallax or maser-orbital kinematics that again probe Friedmann physics, not galactic coupling. (iii) The tier-2 coupling affects *galactic rotation velocities* and *cluster mass budgets*, which are not the observables distance-ladder H_0 probes use. □

Verdict on C. *A line-of-sight functional that accumulates tier-2 weights is well-defined as a mathematical object, but it does not couple to the observables the distance ladder measures.* The v2.2 obstruction stands — now with a sharpened forensic reason: Lemma 1 is the non-trivial content. The original v1 "depth-dependent readout" intuition was reaching for a structure that, even when built cleanly post-V4.5, cannot enter the h-inference pipeline.

3 Hunt for a valid new claim at the V4.5 level

We enumerate the three candidate claims from the forensic brief and evaluate each.

3.1 Candidate 1: tier-1 vs tier-2 narrative reframe

Claim. "τ predicts a single $h = 0.6735$ at tier 1 (cosmic-mean, CMB-scale). Local-ladder probes measure h via tier-2 projections; the observed $\sim$ 0.03–0.06 probe-to-probe spread reflects sector-dependent weighting of the probe's mass distribution, not a true cosmology-scale variation."

Evaluation. Lemma 1 kills the second sentence: local ladders do *not* measure h via tier-2 projections. They measure $d_L(z)$, which is a tier-1 observable in the Friedmann cosmology. Candidate 1 in its current form makes a technical claim that is inconsistent with the two-tier structure as derived by V4.5.

Weaker reformulation. A more defensible phrasing: "τ predicts a single $h = 0.6735$ at tier 1. The probe-to-probe spread observed in local ladders is an *empirical* H_0 tension whose resolution lies in astronomical systematics (Cepheid crowding, TRGB zero-point, etc.), not in tier-2 sector physics." This reformulation is *identical* to the v2.4 posture; it adds no new physics. Candidate 1 reduces to narrative framing.

SH0ES vs TRGB sanity test. SH0ES ($h = 0.730$) and TRGB ($h = 0.698$) both use stellar-sector (D-sector, tier 2 weight 1) calibrators on the distance ladder. Under Candidate 1, both should yield the same h. They disagree at $\sim 3\sigma$. This alone is sufficient to kill any naive "sector-weighting resolves the tension" claim.

3.2 Candidate 2: genuine CMB-vs-local tension

Claim. "The tension is a genuine CMB-vs-local disagreement. τ predicts CMB-side correctly ($\ell_1 = 220.6$, +0.28% of Planck); the local-ladder discrepancy is an *empirical problem* for τ (as it is for ΛCDM). Either a local-ladder systematic is identified, or both frameworks fail at the local-ladder scale."

Evaluation. This is the v2.4 Ledger posture verbatim. It is consistent with V4.5 (tier-1 prediction is correct; tier-2 doesn't affect h) and with the v3 retraction (no τ-native h^{readout} exists). It is honest. It does not claim new physics. It owns the SH0ES/H0LiCOW 3–6σ gap as a shared falsification pressure on both τ and ΛCDM.

Is this a "new" claim? No. It is the current posture. V4.5 confirms it at a deeper structural level (we now know *why* tier-2 weights can't fix the tension, rather than just observing that depth-dependent readouts don't derive). Candidate 2 is the verdict-(b) outcome.

3.3 Candidate 3: tier-2 probe-sector-weighting formula

Claim. "There exists a V4.5-derived tier-2 projection formula $h^{\text{tier}-2}(\text{probe})$ that could reproduce observed probe-dependent h-values if the sector-weighting of each probe is computed correctly."

This is the ambitious claim; we attempt its derivation in §4. We already have reason to doubt from Lemma 1, but the derivation is worth attempting to see whether any non-trivial probe-dependent prediction emerges.

4 Attempted Candidate 3 derivation

4.1 Probe sector taxonomy

We classify each H_0 probe by its tier-2 sector — i.e., which matter phase dominates the physical observable.

Probe	Physical observable	Phase	Tier-2 sector	Weight
Planck CMB	angular acoustic peak	photon-baryon fluid	tier-1 (global)	—
Cepheid (SH0ES)	stellar pulsation P-L	star (D)	D	1
TRGB	RGB-tip luminosity	star (D)	D	1
JAGB	AGB luminosity	star (D)	D	1
SN Ia (in SH0ES)	thermonuclear WD	WD (compact-star, D)	D	1
Standard sirens	BH/NS merger GW	compact object (D)	D	1
Megamasers	maser in AGN disk	warm molecular gas	ω or mix	κ_ω or mix
H0LiCOW lensing	time-delay integral	mixed star+gas+DM	tier-1 (bg.)	—

Immediate problem. Five of the seven probes (Cepheid, TRGB, JAGB, SN Ia, sirens) sit in the *same* tier-2 sector (D, weight 1). Under any formula that is a function of tier-2 sector alone, they should predict the *same* h. They do not:

$$h_{\text{Cepheid}} = 0.730, \quad h_{\text{TRGB}} = 0.698, \quad h_{\text{JAGB}} = 0.695, \quad h_{\text{siren}} = 0.700.$$

The spread among same-sector probes is $\Delta h \sim 0.035$, nearly the full tension we were hoping to explain.

Rescue 1: sub-sector refinement. Could the D-sector split into further sub-sectors (main-sequence vs giant-branch vs compact-object)? Within the V4.5 framework, the tier-2 sector assignment uses the decoherence-rate criterion Γ_{decoh}/H applied to local matter phases. For stars (any type), the luminous envelope has $\Gamma_{\text{decoh}}/H \ll 1$ (optically thick stellar atmospheres at long phase-coherence times); all stars sit deep in the D-sector by this criterion. No sub-sector structure is derivable from the V4.5 decoherence-rate table.

Rescue 2: probe-dependent accumulation. Could the probe yield a *line-of-sight-integrated* tier-2 weight that differs across distance to the source, even within the same sector? For example, a long Cepheid ladder (many rungs in spiral galaxies with different gas fractions) vs a short TRGB ladder (few rungs, elliptical hosts with less gas). This would be a *line-of-sight functional*, which Lemma 1 ruled out: the distance-ladder calibration depends on stellar physics (calibrated from LMC, NGC 4258, etc.), not on an accumulation of galactic sector weights. The inferred h is a function of $d_L(z)$, not of what galaxies sit along the LOS.

Rescue 3: SN Ia host-galaxy sector. Could SN Ia calibration in SH0ES be affected by the host galaxy's gas fraction, via a host-mass step? This is a *known empirical effect* in SN cosmology (the "mass step"), but its origin is orthodox astrophysics (metallicity, progenitor mass, environment). It is a *systematic*, not a tier-2 effect of the τ framework. Framing it as tier-2 would be the exact "semantic maneuver" that R4 (McGaugh) flagged in v2.2.

Verdict on Candidate 3. *Candidate 3 fails.* Five of the seven probes live in the same tier-2 sector and already disagree at $\sim 3\sigma$; the tier-2 structure cannot discriminate among them. The remaining two (megamasers, lensing) are either consistent with the τ prediction ($h_{\text{maser}} = 0.720 \pm 0.030$, 1.6σ) or in tension ($h_{\text{H0LiCOW}} = 0.733$, 3.3σ) at levels that match the orthodox H_0 tension pattern, without needing tier-2 physics.

Verdict (F1.main)**.** *The V4.5 two-tier coherence criterion sharpens the forensic reasons why Approaches A/B/C all failed (A: tiers do not encode z; B: H_0 is tier-1, the observable is upstream of tier-2 projection; C: ladders probe tier-1 Friedmann physics, not tier-2 local weights). It does* not *produce a new falsifiable h-prediction. Candidate 1 reduces to narrative framing; Candidate 2 is the current v2.4 posture; Candidate 3 fails because five of seven probes share the same tier-2 sector yet disagree at 3σ, and the remaining two (megamasers, H0LiCOW) do not need tier-2 physics to match or to miss.*

The v2.2 retraction ***stands****. The correct v2.5 update is verdict (b):* ***incorporate the two-tier insight as a structural clarification of why no*** h^{readout} ***exists, not as new physics.***

5 Implications for v2.5

5.1 Narrative refinement, not new claim

The v2.5 Physics Ledger should refine the Ch. 63 retraction note (`rem:ch63-h0-depth-flag`) and the Ch. 68 N21 annotation to *include the two-tier structural explanation*, phrased as a forensic strengthening of the v2.2 retraction, not as a rescue:

> *The orbit-depth-dependent readout is retracted at v2.2; the retraction is forensically strengthened at v2.5 via the V4.5 two-tier coherence criterion. Under the criterion, H_0 is a tier-1 observable (an expansion rate of the cosmic-mean background), and tier-2 local matter-phase couplings $\{1, \kappa_\omega, \approx 0\}$ do not propagate into it. The distance ladder probes Friedmann $H(z)$, not galactic sector weights. Hence even the structurally refined framework post-V4.5 does not admit a $h^{\text{readout}}(z_{\text{probe}}, \text{tier} - 2 \text{ sector})$ formula. The single-value τ prediction $h = 0.6735$ is forced.*

5.2 Proposed N21 phrasing (v2.5)

> ***N21 (v2.5):*** *τ predicts $h = 2/3 + \iota_\tau^2/17 = 0.6735$ as the unique cosmological Hubble parameter, computed from the tier-1 D-sector holonomy ratio on $H_\partial[\omega]$ (V.T198, M3h). The tier-1/tier-2 decomposition (V4.5) prevents any tier-2 sector-weight modification of this prediction: H_0 is upstream of the trinary decomposition. The v1 "depth-dependent readout" is retracted with the forensic reason that no tier-2 projection can reach the h-observable.*
>
> ***Consistency:*** *Planck (*0.6735, 0.0σ*), standard sirens (*0.700, 0.7σ*), megamasers (*0.720, 1.6σ*), JAGB (*0.695, 1.7σ*).*
>
> ***Tension:*** *TRGB (*0.698, 2.7σ*), H0LiCOW (*0.733, 3.3σ*), SH0ES (*0.730, 5.6σ*).*
>
> ***Falsifier (unchanged from v2.2):*** *If the community H_0 concordance converges above* 0.70 *with SH0ES/TRGB/H0LiCOW systematics resolved, τ's h-prediction is falsified. The two-tier clarification closes the "extend the readout signature" escape hatch that was still formally available at v2.2: no such extension is compatible with tier-1 placement of H_0.*

5.3 What this does not foreclose

Two honest research avenues remain open, neither of which reopens Approach A/B/C:

1. **A tier-1 shift.** If a future derivation modifies the tier-1 M3h holonomy ratio (changing $(1-\iota_\tau)/\iota_\tau^2$ to a different functional form), $h = 0.6735$ itself could shift. This would be a tier-1 revision, not a readout extension, and would require re-deriving $\ell_1 = 220.6$ jointly. The +0.28% agreement with Planck makes this route very constrained: any tier-1 change that preserves ℓ_1 constrains h to within $\sim 0.5\%$ of the current value.

2. **A tier-2 empirical anchor for the Bullet Cluster.** Independent of h, the V4.5 verdict that cluster mass budgets are sourced by the tier-1 D-sector background rather than by tier-2 ICM coupling remains to be tested against the Bullet Cluster offset (v3 target). This is unrelated to h^{readout}; we flag it only to note that tier-2 structure has other open predictions.

6 Summary table and honest posture

	v2.2 obstruction	V4.5 reframe	Status
Approach A (scalar $f(z)$)	data not single-valued	tiers orthogonal to z	dead
Approach B (readout functor)	no depth argument	h is tier-1, upstream	dead
Approach C (LOS functional)	no LOS on $H_\partial[\omega]$	ladders are tier-1 (Lemma 2)	dead
Candidate 1 (narrative)	—	reduces to v2.4 posture	not a new claim
Candidate 2 (honest tension)	—	current posture, V4.5-strengthened	current v2.4
Candidate 3 (tier-2 formula)	—	5-of-7 in same sector, 3σ gap	dead

Observation. *Every approach that attempts to derive a τ-native h^{readout} fails, and the V4.5 two-tier criterion makes the failure* structural *rather than merely* signature-level*: H_0 is a tier-1 observable and cannot be reached by tier-2 projections. The retraction stands; the two-tier insight is narrative refinement for v2.5.*

7 Final verdict

Verdict (F1.final)**.** ***Outcome (b) of the charge.*** *The two-tier structure refines the H_0-tension* narrative *but does* not *produce a new falsifiable prediction. The v2.2 retraction of "depth-dependent readout" stands. V2.5 should incorporate the two-tier insight as a structural clarification of the retraction's forensic basis, not as a rescue. The N21 falsifier is tightened: the "extend the readout signature" escape hatch available at v2.2 is now closed by V4.5, since no extension compatible with tier-1 placement of H_0 exists. The single τ-native prediction $h = 0.6735$ is the Ledger's final H_0 claim, and the 3–6σ tension with TRGB/H0LiCOW/SH0ES is a genuine empirical test.*

Research discipline remark. This forensic looked for what is *actually there* given V4.5, not for a convenient rescue. The V4.5 insight is real and valuable (it resolves the v2.3.1 cosmology-consistency flag), but it does *not* happen to supply the missing piece of infrastructure a depth-dependent readout would have needed. Reporting that cleanly is the point; inventing a fifth approach would be rescue physics.